

DS Unit 2 — Stacks & Queues Complete Notes

Data Structures (DS) · Unit 2 · Stacks & Queues · 2025

Overview

These notes cover **Stacks & Queues**. Core data structures and the algorithms built on them — arrays, lists, stacks, trees, graphs, sorting and hashing. Each section states the core idea, a worked intuition and the points examiners look for.

1. Stack ADT

A stack is a LIFO structure supporting push, pop and peek in constant time.

Why it matters: Stack ADT appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of stack adt in one line.
- A small example showing stack adt in practice.
- Common mistakes students make and how to avoid them.

2. Infix to Postfix Conversion

Operator precedence and a stack convert infix expressions to postfix for easy evaluation.

Why it matters: Infix to Postfix Conversion appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of infix to postfix conversion in one line.
- A small example showing infix to postfix conversion in practice.
- Common mistakes students make and how to avoid them.

3. Recursion and the Call Stack

Each recursive call pushes a frame; tail recursion can be converted to iteration.

Why it matters: Recursion and the Call Stack appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of recursion and the call stack in one line.
- A small example showing recursion and the call stack in practice.
- Common mistakes students make and how to avoid them.

4. Queue ADT

A queue is FIFO; circular arrays avoid wasting space after dequeues.

Why it matters: Queue ADT appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of queue adt in one line.
- A small example showing queue adt in practice.
- Common mistakes students make and how to avoid them.

5. Deque and Priority Queue

Dequeues allow insertion and deletion at both ends; priority queues serve the highest-priority element first.

Why it matters: Deque and Priority Queue appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of deque and priority queue in one line.
- A small example showing deque and priority queue in practice.
- Common mistakes students make and how to avoid them.

6. Applications of Stacks and Queues

Balanced parentheses, undo operations, BFS and CPU scheduling all rely on stacks or queues.

Why it matters: Applications of Stacks and Queues appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of applications of stacks and queues in one line.
- A small example showing applications of stacks and queues in practice.
- Common mistakes students make and how to avoid them.

Quick revision

Stack ADT · Infix to Postfix Conversion · Recursion and the Call Stack · Queue ADT · Deque and Priority Queue · Applications of Stacks and Queues